

PPUS — Prediction Probability & Uncertainty Standard

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Operational Layer: Prediction Probability & Uncertainty Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard

Governed Space:

Predictive Probability & Uncertainty

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AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Predictive Probability & Uncertainty

Canonical Definition

Prediction Probability & Uncertainty Standard (PPUS) defines the governance conditions under which probability, uncertainty, confidence, ranges, intervals, conditionality, distributional representation, and related predictive qualification are associated with a Governed Prediction without overstating what the prediction actually establishes.

PPUS governs how uncertainty about future outcomes is represented after a specific Predictive Claim and Governed Prediction Output have been formed.

It ensures that statements such as:



70% probability

high confidence

likely

10–15% range

low uncertainty

or:

expected outcome

remain tied to a documented predictive meaning, methodology, applicable conditions, and representational boundary.

PPUS does not make a prediction more certain.

It governs how much certainty, probability, confidence, uncertainty, and conditionality the prediction is legitimately permitted to express.

The canonical transition is:

Governed Prediction + Formation Trace + Applicable Probability / Uncertainty Methodology → Governed Probability & Uncertainty Representation

Core Question

How should probability, uncertainty, confidence, ranges, and conditionality be represented without overstating what the prediction actually establishes?

Foundational Principle

Probability is not certainty, confidence is not probability, and uncertainty does not disappear because a prediction has been quantified.

Operational Role

PPUS serves as the sixth governed space of the Prediction Governance Architecture (PGA™) .

It receives a prediction that has already passed through:

Prediction Object & Purpose Governance; Prediction Methodology Governance; Prediction Basis Governance; Prediction Assumption Governance; Prediction Formation Governance.

Its role is to govern the representational layer between:

what is predicted

and:

how strongly the prediction may legitimately be expressed.

PPUS establishes the conditions under which a prediction may carry:

probability; probability range; confidence; uncertainty; confidence interval; predictive interval; distribution; qualitative likelihood; conditional probability; scenario-dependent probability; probability provenance; uncertainty qualification.

It prevents:

numerical expression

from being mistaken for:

epistemic certainty.

Position Within PGA™

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Prediction Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Current Position: Prediction Probability & Uncertainty

Probability Governance

A probability associated with a prediction must possess governed meaning.

A probability value should not stand alone.

Its governance should establish, where applicable:

what event or state it applies to; what Prediction Horizon applies;

what conditions apply; how it was produced; which methodology supports it; what population, system, scenario, or boundary applies; whether it is point probability or range; whether it is conditional; how uncertainty around the probability itself is represented.

Probability ≠ Certainty

A probability value describes a predictive relationship.

It does not establish certainty about future reality.

99% Probability ≠ Guaranteed Outcome

Likewise:

1% Probability ≠ Impossible Outcome

Probability ≠ Confidence

Probability and confidence answer different governance questions.

Probability concerns:

How likely is the predicted event or state according to the applicable predictive methodology?

Confidence concerns:

How strongly may the predictive representation itself be relied upon as a representation of that estimated probability or predictive conclusion?

A prediction may therefore have:

high estimated probability + low confidence

or:

moderate probability + high confidence in the estimate.

Confidence ≠ Certainty

Confidence does not convert a prediction into fact.

A highly governed confidence state may indicate strong methodological support.

It does not establish future truth.

Probability Provenance

Every probability should remain traceably associated with its source.

Probability provenance may include:

statistical estimation; model output; expert elicitation; Bayesian update; frequency-based estimate; simulation-derived estimate; market-implied probability; collective forecast; transformed score; hybrid methodology.

Same Probability Value $\neq$ Same Predictive Meaning

Two predictions may both state:

70%

while meaning materially different things.

Probability Source

The origin of probability affects interpretation.

Example:

70% statistical model probability

is not automatically equivalent to:

70% expert confidence

or:

70% market-implied probability.

The numeric value alone does not create equivalence.

Market-Implied Probability

Where a prediction market produces an implied probability, PPUS governs how that probability is represented if incorporated into PGA™.

Market-Implied Probability $\neq$ Governed Probability Estimate by Default

A market price may reflect:

liquidity; participant composition; incentives; information asymmetry; trading strategy; market rules; manipulation; hedging; capital constraints.

The value may still be useful.

Its meaning must remain governed.

Probability Representation Requires Methodological Context

A probability should remain linked to the methodology that produced or justified it.

Probability Without Methodological Context $\neq$ Fully Governed Probability

Point Probability

A Point Probability expresses one numeric probability estimate.

Example:

Probability of Event X: 72%

A point estimate may conceal uncertainty about the probability itself.

PPUS therefore distinguishes:

probability estimate

from:

certainty about the probability estimate.

Probability Range

A Probability Range expresses probability within an interval.

Example:

Estimated probability: 65–75%

A range may better reflect:

model uncertainty; parameter uncertainty; basis uncertainty; assumption sensitivity; methodological disagreement.

Probability Distribution

Where appropriate, a predictive system may represent a distribution rather than a point or simple range.

The distribution should remain linked to:

predicted variable; methodology; conditions; horizon; applicable assumptions.

Distribution Shape

Where material, the representation may preserve whether the predictive distribution is:

symmetric; skewed; multimodal; heavy-tailed; truncated; discrete; continuous.

A mean alone may not sufficiently represent predictive uncertainty.

Expected Value

An Expected Value represents a mathematical expectation under a probability distribution.

It must not automatically be interpreted as:

the most likely outcome

or:

the outcome that will occur.

Expected Value $\neq$ Most Probable Outcome

Most Likely Outcome

The most likely outcome may differ from:

mean; median; expected value; central range.

These should not be silently treated as equivalent.

Median Prediction

A median predictive value represents the point at which the relevant predictive distribution is divided according to the applicable methodology.

It does not automatically represent the most likely outcome.

Mode

Where relevant, the modal value represents the most probable outcome or range under the specified distribution.

Mean $\neq$ Median $\neq$ Mode

Uncertainty

Predictive Uncertainty represents governed uncertainty concerning the future state, event, magnitude, timing, probability, or applicability of a prediction.

Uncertainty may arise from:

incomplete information; measurement limitations; model limitations; parameter uncertainty; assumption uncertainty; environmental change; structural uncertainty; scenario uncertainty; behavioral uncertainty; temporal uncertainty;

boundary uncertainty; methodological disagreement; unknown dependencies.

Uncertainty ≠ Error

Uncertainty exists before the outcome is known.

Prediction Error is determined later through outcome comparison.

Predictive Uncertainty ≠ Observed Prediction Error

Uncertainty ≠ Ignorance by Default

Some uncertainty can be quantified.

Some can only be qualified.

Some may remain unknown.

PPUS permits multiple forms of uncertainty representation.

Known Uncertainty

Known uncertainty can be identified and represented.

Example:

Sensor measurement $\pm 2\%$.

Partially Quantifiable Uncertainty

Some uncertainty can be bounded but not precisely measured.

Example:

Model sensitivity indicates demand forecast may shift materially if interest rates exceed a defined threshold.

Unquantified Uncertainty

Where uncertainty cannot be responsibly quantified, PPUS permits qualitative representation.

Examples:

High structural uncertainty

Unknown regulatory sensitivity

Material but unquantified supply-chain uncertainty

Unquantified ≠ Nonexistent

Unknown Uncertainty

Some uncertainty may not yet be known.

PPUS cannot require governance of an unknown unknown as though it were already identifiable.

However, methodologies should avoid representing:

absence of identified uncertainty

as:

absence of uncertainty.

Aleatory Uncertainty

Where relevant, implementations may distinguish uncertainty arising from inherent variability.

Example:

random environmental variation.

Epistemic Uncertainty

Where relevant, implementations may distinguish uncertainty arising from incomplete knowledge.

Example:

unknown model parameter.

PPUS does not mandate these categories universally but permits governed distinction.

Structural Uncertainty

Structural Uncertainty concerns uncertainty about whether the predictive model or conceptual relationship adequately represents the governed predictive environment.

This may be more consequential than parameter variation.

Model Uncertainty

Multiple plausible predictive models may produce different outputs.

PPUS may govern how that disagreement is represented.

One Selected Model $\neq$ No Model Uncertainty

Parameter Uncertainty

Model parameters may themselves be uncertain.

Where material, the probability representation should not imply greater precision than parameter uncertainty permits.

Input Uncertainty

Prediction Basis elements may contain uncertainty.

PBAS governs the basis.

PPUS governs how such uncertainty propagates into predictive representation where applicable.

Assumption Uncertainty

PAS governs assumptions and their state.

PPUS governs how uncertainty associated with those assumptions affects the probability or uncertainty representation of the prediction.

Horizon Uncertainty

Predictive uncertainty may increase with Prediction Horizon.

A probability estimated for:

7 days

may not remain meaningful for:

12 months.

Probability at Horizon A $\neq$ Probability at Horizon B

Boundary Uncertainty

Prediction uncertainty may vary across geographic, organizational, system, or population boundaries.

A national forecast may conceal substantial regional uncertainty.

Conditional Probability

A Conditional Probability represents probability under specified conditions.

Canonical form:

$P(\text{Event X} \mid \text{Condition Y})$

The condition is part of the predictive meaning.

Conditional Probability Without Condition $\neq$ Same Probability

Condition Removal

Removing a condition may materially alter probability.

Example:

80% probability of completion if funding remains available

cannot become:

80% probability of completion

without changing the prediction.

Scenario-Dependent Probability

Probability may differ across scenarios.

Example:

Base scenario: 65%

Adverse scenario: 32%

Favorable scenario: 81%

These are distinct conditional predictive representations.

Probability Across Assumption States

A prediction may have different probability under different assumption states.

Example:

Supplier continuity Holding:

78%

Supplier continuity Under Challenge:

46%

PPUS preserves the conditional relationship.

Confidence

Confidence represents the governed strength with which a probability estimate, predictive range, or predictive representation may be treated as supported under the applicable methodology and conditions.

Confidence may be:

quantitative; qualitative; methodology-specific.

Confidence Level

Where statistical methodology supports confidence levels, the meaning of that level must remain explicit.

A:

95% confidence interval

does not mean:

95% probability that the true fixed parameter is inside this realized interval

under all statistical frameworks.

PPUS requires the relevant methodological interpretation to remain attached.

Confidence Label

Qualitative labels may include:

Low

Moderate

High

where their meaning is defined.

Qualitative labels must not masquerade as universal mathematical quantities.

High Confidence + Low Probability

Example:

A well-supported model may estimate:

5% probability of failure

with:

high confidence in the estimate.

This is conceptually legitimate.

Low Confidence + High Probability

A weak basis may produce:

80% estimated probability

while confidence in that estimate remains low.

This should not be represented as a strong prediction merely because the probability number is high.

Probability–Confidence Matrix

Implementations may represent probability and confidence separately.

For example:

Probability Confidence Interpretation High High High estimated likelihood with strong support High Low
High estimated likelihood with weak support Low High Low estimated likelihood with strong support Low
Low Low estimated likelihood with weak support

The categories must be methodologically defined.

Prediction Range

A prediction may be expressed through:

point estimate; interval; percentile range; scenario range; tolerance range; confidence interval; predictive interval.

PPUS requires the type of range to remain explicit.

Range ≠ Uncertainty by Default

A range may represent:

model uncertainty; expected variability; scenario bounds; target band; confidence interval.

These are not interchangeable.

Confidence Interval

A Confidence Interval is a statistical construct whose interpretation depends upon the methodology used.

It should not be relabeled as:

future outcome range

unless that interpretation is methodologically justified.

Predictive Interval

A Predictive Interval concerns a future observed value or set of values under the applicable predictive methodology.

It is distinct from a parameter confidence interval.

Confidence Interval ≠ Predictive Interval

Credible Interval

Where Bayesian methodology is used, a Credible Interval may have different interpretive semantics from a frequentist confidence interval.

PPUS preserves methodological provenance.

Tolerance Interval

Tolerance intervals may represent population coverage rather than predictive probability in the same sense as a forecast range.

They should remain properly identified.

Range Width

A narrow range may appear more useful.

It is not automatically better.

Narrower Range ≠ Better Prediction by Default

A narrow range that excludes plausible uncertainty may represent overconfidence.

Wide Range

A wide range may honestly represent substantial uncertainty.

Wide Range $\neq$ Poor Governance by Default

False Precision

PPUS prohibits representational precision exceeding methodological support.

Examples:

67.438%

when the basis only supports an approximate probability range.

More digits can create the appearance of certainty without adding legitimate predictive information.

Numerical Precision $\neq$ Epistemic Precision

Probability Calibration

Calibration concerns whether stated probability levels correspond appropriately with observed outcome frequencies over a governed set of predictions.

Example:

Among predictions assigned approximately 70% probability, approximately 70% of governed outcomes may be expected to occur under an appropriately calibrated system and evaluation structure.

POFPS later evaluates actual calibration performance.

PPUS governs how probability representations remain suitable for such later assessment.

Calibration $\neq$ Individual Prediction Correctness

A well-calibrated system may produce individual predictions that are wrong.

Calibration $\neq$ Certainty

Probability Resolution

A system may distinguish effectively between low- and high-likelihood events.

Resolution is not identical to calibration.

PPUS preserves the possibility of later performance analysis without prematurely determining it.

Probability Discrimination

Where applicable, a predictive system may distinguish relative likelihood among outcomes.

This differs from absolute calibration.

Qualitative Likelihood

Where numerical probability is not justified, predictive systems may use qualitative likelihood terms.

Examples:

Rare

Unlikely

Possible

Likely

Highly Likely

These terms require governed definitions if they materially affect interpretation or reliance.

Likely ≠ Numerically Defined Probability by Default

The word:

likely

must not automatically be interpreted as a universal numeric threshold unless the applicable framework defines it.

Cross-System Likelihood Labels

System A may define:

Likely = 60–80%

while System B defines:

Likely = 70–90%.

The same label therefore may not carry the same predictive meaning.

Same Likelihood Label ≠ Same Probability Meaning

Probability Category Mapping

Where numeric probabilities are mapped into categories, thresholds must remain governed.

Example:

0–20% = Low

21–60% = Medium

61–100% = High

The mapping should remain identifiable.

Threshold Sensitivity

Small numeric changes near category thresholds may cause large semantic changes.

Example:

59.9% = Medium

60.1% = High

PPUS should preserve the underlying number where material.

Rounded Probability

Rounding must not create misleading interpretation.

49.6% → 50%

may matter where 50% carries a decision or communication threshold.

Probability Transformation

Probability may be transformed from:

score; odds; log-odds; model confidence; market price; ranking; ensemble output.

The transformation must remain governed.

Score $\neq$ Probability by Default

Odds $\neq$ Probability

Odds can be converted into probability, but they are not the same representation.

Model Confidence $\neq$ Event Probability

Some AI systems output a confidence score concerning classification or model certainty.

That score must not automatically be presented as the probability that a real-world event will occur.

Classification Confidence $\neq$ Predictive Probability

This distinction is especially important in machine learning systems.

Market Price $\neq$ Governed Probability Estimate

A market price may be converted into an implied probability.

It remains a market-derived signal unless sufficient governance establishes the conditions under which it may be treated as a probability estimate.

Collective Probability

Where multiple predictors provide probabilities, CPGS governs collective composition.

PPUS governs the probability semantics of the resulting Governed Prediction.

Probability Aggregation

Averaging several probabilities does not automatically create a legitimate probability.

Average of Probabilities $\neq$ Governed Collective Probability by Default

Dependent Predictors

Where predictive sources are dependent, apparent agreement may overstate confidence.

Agreement Among Dependent Predictors $\neq$ Independent Confirmation

Uncertainty Propagation

Where Prediction Basis, assumptions, methodology, or transformations contain uncertainty, PPUS should permit that uncertainty to propagate into the final predictive representation where material.

Ignoring upstream uncertainty may create artificial certainty.

Uncertainty Compression

A complex predictive process may collapse many uncertainties into one number.

This may be operationally useful.

But material uncertainty must not disappear merely because the final interface is simple.

Simple Output $\neq$ Simple Uncertainty Structure

Uncertainty Decomposition

Implementations may decompose uncertainty into categories such as:

basis uncertainty; assumption uncertainty; model uncertainty; parameter uncertainty; scenario uncertainty; timing uncertainty; boundary uncertainty.

This may improve transparency.

Dominant Uncertainty Source

Where one uncertainty source materially dominates the prediction, it may require explicit representation.

Example:

A revenue forecast may have narrow statistical variance but high regulatory uncertainty.

Uncertainty Concentration

Multiple uncertainties may depend upon one shared condition.

A prediction can therefore appear diversified while uncertainty remains concentrated.

Uncertainty Interaction

Uncertainty sources may interact nonlinearly.

Example:

supplier disruption + exchange-rate shock + demand volatility.

PPUS should permit compound uncertainty representation where material.

Uncertainty Amplification

Small upstream uncertainty may produce large predictive uncertainty.

Uncertainty Dampening

A prediction may remain relatively stable despite uncertain inputs.

PPUS distinguishes:

uncertainty exists

from:

uncertainty materially affects prediction.

Confidence Source

Confidence should remain associated with what supports it.

Possible sources include:

statistical estimation; historical validation; model robustness; basis quality; assumption stability; convergence of independent methods; domain expertise.

Confidence should not be assigned merely because a model generates a numeric probability.

Confidence Overstatement

A prediction should not be labeled:

High Confidence

when the actual governance basis is:

weak; unknown; highly conditional; methodologically contested.

Confidence Understatement

Governance should also avoid unnecessary understatement where strong support exists.

PPUS seeks faithful representation, not systematically conservative representation.

Probability and Feasibility

Probability and feasibility are distinct.

PFES governs whether the predicted outcome is realistically achievable under the relevant conditions.

PPUS governs estimated likelihood and uncertainty.

Probable $\neq$ Feasible

Feasible $\neq$ Probable

Example:

A theoretically feasible event may have extremely low probability.

An event estimated as probable may later be found infeasible because an operational constraint was omitted.

Probability and Prediction Basis

PBAS governs what information legitimately forms Prediction Basis.

PPUS governs how uncertainty in or about that basis affects predictive probability representation.

Strong Prediction Basis $\neq$ High Probability

Probability and Assumptions

PAS governs assumptions.

PPUS governs probability conditional upon those assumptions where applicable.

Strong Assumption Support $\neq$ High Outcome Probability

Probability and Prediction Formation

PFOS establishes the specific Predictive Claim.

PPUS qualifies that claim with probability, uncertainty, confidence, and related representation.

Prediction Formation $\neq$ Probability Determination

Probability and Communication

PCRS later governs how probability may be communicated and relied upon.

PPUS establishes the governed probability meaning before communication occurs.

Probability and Revision

PMRS may later require probability revision when:

information changes; assumptions change; conditions change; methodology changes.

PPUS governs the representation applicable to each prediction state.

Current Probability $\neq$ Historical Probability

Later probability updates must not overwrite the probability state attached to the original prediction version.

Probability Change

Probability may change even where the Predictive Claim remains identical.

Example:

Same claim:

Project completes by 31 December 2026.

Initial probability:

65%

Later probability:

40%

The claim may remain the same while predictive qualification changes.

Probability Change $\neq$ New Predictive Claim by Default

Whether a new prediction version is required depends upon the broader lifecycle governance under PMRS and PRCS.

Continuing Probability Validity

A probability estimate may become stale as predictive conditions change.

Valid Initial Probability $\neq$ Valid Continuing Probability

Prediction Basis Decay

A prediction's basis may lose relevance over time.

This does not automatically invalidate the prediction.

However, it may reduce confidence, increase uncertainty, or require review.

Historical Predictive Conditions $\neq$ Current Predictive Conditions

PPUS preserves the distinction between probability originally assigned and probability currently supportable.

Unknown Probability

A Governed Prediction may legitimately carry:

Probability Unable to Determine

where no defensible probability estimate exists.

This is preferable to inventing a number.

Unknown Probability $\neq$ 50% Probability

50% $\neq$ Unknown

A 50% probability is a specific predictive estimate.

It should not be used as a substitute for lack of knowledge.

No Probability Assigned

Some governed predictions may not require numerical probability.

Example:

a scenario-conditioned strategic prediction may use qualitative uncertainty instead.

PPUS does not require numerical probability for every prediction.

Quantification Requirement

Probability should be quantified only where the applicable methodology supports meaningful quantification.

Quantifiable $\neq$ Must Be Quantified

Probability Governance Gap

A Probability Governance Gap exists where probability representation cannot be adequately associated with:

the claim; methodology; horizon; conditions; basis; assumptions; provenance.

Uncertainty Governance Gap

An Uncertainty Governance Gap exists where material uncertainty is:

unknown; unrepresented; incorrectly quantified; improperly compressed; misleadingly omitted.

Confidence Governance Gap

A Confidence Governance Gap exists where confidence representation lacks sufficient support or meaning.

Representation Governance Gap

A broader gap exists where:

probability; confidence; uncertainty; range; likelihood label
could reasonably be interpreted in materially different ways.

Representation States

A probability or uncertainty representation may occupy states such as:

Established

Conditionally Established

Limited

Uncertain

Conflicting

Insufficiently Supported

Unable to Determine

Multiple Probability Estimates

A single Predictive Claim may receive multiple probability estimates from different methodologies.

PPUS requires their provenance and relationship to remain identifiable.

It does not force them into one number.

Probability Disagreement

Disagreement between legitimate methods may itself be informative.

Probability Disagreement $\neq$ Governance Failure by Default

Methodological Disagreement

Example:

Model A:

65%

Expert panel:

35%

Market-implied:

58%

PPUS preserves the distinct origins.

CPGS governs collective composition where they are combined into a collective prediction.

Probability Consensus

Agreement across independent methods may strengthen confidence.

But independence must not be assumed.

Probability Agreement $\neq$ Independent Confirmation by Default

High Probability / Low Evidence

A high probability generated from weak basis remains high numerically but may deserve low confidence.

PPUS preserves both dimensions.

Low Probability / High Consequence

Probability does not determine consequence.

Risk governance belongs elsewhere where applicable.

Low Probability $\neq$ Low Consequence

Probability Semantics in AI Systems

AI systems often emit values labeled:

confidence; likelihood; score; probability.

PPUS requires each label to have defined semantics.

A softmax score, classifier confidence, anomaly score, or ranking score cannot automatically become a real-world event probability.

Score-to-Probability Conversion

Where a score is transformed into probability, the transformation methodology must remain governed.

Uncertainty in AI-Generated Predictions

AI-generated predictions may contain uncertainty from:

training distribution; model architecture; data shift; hidden state; prompt/context sensitivity; stochastic execution; model calibration; downstream orchestration.

PPUS governs representation of those uncertainties where they materially affect the prediction.

Predictive Distribution Shift

Where the operational environment differs materially from the environment supporting probability estimation, confidence may decline or uncertainty may increase.

ORA may establish relevant Operational Reality signals.

PPUS governs their effect on predictive qualification.

Probability Governance and Operational Reality

PPUS does not determine Operational Reality.

ORA retains that space.

PPUS may consume governed Operational Reality references where they affect:

conditions; uncertainty; probability validity; confidence.

Probability Governance and Evidence

ADIT® governs evidence and audit-related evidence structures.

PPUS does not independently validate evidence.

Probability Governance and Validation

VALIDOS™ may validate relevant structures or claims under its own governed domain.

PPUS owns predictive probability and uncertainty representation.

Probability Governance and Integrity

INTEGROS® may govern integrity structures.

PPUS does not absorb general integrity governance.

Probability Governance and Accountability

AGA™ governs accountability.

PPUS does not determine who is accountable for a probability estimate.

Probability Governance and Authority

GOA™ governs authority structures.

PPUS does not determine who has authority to publish or rely upon the prediction.

Probability Governance Requirements

PPUS requires that:

1. probability remain associated with a specific Predictive Claim;
2. probability meaning remain explicit;
3. probability provenance remain identifiable;
4. probability methodology remain traceable;
5. Prediction Horizon remain associated with probability;
6. Predictive Boundary remain associated where material;
7. conditions remain associated with conditional probability;
8. scenario-dependent probabilities remain distinguishable;
9. probability remain distinguishable from certainty;
10. probability remain distinguishable from confidence;
11. confidence remain distinguishable from certainty;
12. point probabilities not conceal material uncertainty where qualification is necessary;
13. probability ranges identify their intended meaning;
14. probability distributions preserve applicable semantics;
15. expected value remain distinguishable from most likely outcome;
16. mean, median, and mode remain distinguishable where material;
17. material uncertainty sources remain identifiable where possible;
18. unquantified uncertainty remain representable;
19. unknown uncertainty not be represented as absent uncertainty;
20. confidence representations possess governed meaning;
21. confidence labels not be treated as universal numeric

probabilities by default; 22. confidence intervals remain distinguishable from predictive intervals; 23. Bayesian credible intervals remain distinguishable where methodology requires; 24. ranges not imply greater certainty than supported; 25. false precision be avoided; 26. rounding not materially alter probability meaning; 27. qualitative likelihood terms possess governed semantics where material; 28. scores remain distinguishable from probabilities; 29. classifier confidence remain distinguishable from real-world event probability; 30. market-implied probability remain distinguishable from governed probability estimate by default; 31. multiple probability estimates preserve source and methodology; 32. dependent predictive sources not be treated as independent confirmation by default; 33. upstream uncertainty be capable of propagation into predictive representation; 34. major uncertainty concentration remain visible where material; 35. probability and feasibility remain separate governance concepts; 36. probability and consequence remain separate; 37. inability to assign probability remain a legitimate result; 38. unknown probability not default to 50%; 39. historical probability states remain reconstructable; 40. probability changes not silently rewrite earlier prediction states.

Governance Logic

PPUS applies the following governance logic:

Governed Prediction

Governed Prediction Formation Trace

↓

Probability Requirement Determined

↓

Probability Source / Methodology Identified

↓

Probability Value / Range / Distribution Established where supportable

↓

Prediction Horizon / Boundary / Conditions Associated

↓

Probability Provenance Preserved

↓

Uncertainty Sources Identified

↓

Quantifiable / Partially Quantifiable / Unquantified Uncertainty Distinguished

↓

Confidence Representation Established where applicable

↓

Probability / Confidence / Uncertainty Semantics Separated

↓

Range / Interval Type Identified

↓

False Precision / Rounding / Transformation Risks Assessed

↓

Conditional / Scenario-Dependent Representation Preserved

↓

Representation Gaps Assessed

↓

Historical Probability & Uncertainty State Preserved

↓

Governed Probability & Uncertainty Representation

↓

PFES — Prediction Feasibility Standard

Where sufficient governance cannot be established:

Probability / Uncertainty Governance Gap → Clarification / Qualification / Reduced Precision / Conditional Representation / Uncertainty Disclosure / Probability Unable to Determine

Governance Outputs

PPUS may produce:

Governed Probability Representation; Governed Uncertainty Representation; Probability Value; Probability Range; Probability Distribution; Probability Provenance Record; Probability Methodology Reference; Probability Horizon Reference; Conditional Probability Record; Scenario Probability Record; Confidence Representation; Confidence State; Uncertainty Register; Uncertainty Source Map; Uncertainty Range; Uncertainty Qualification; Predictive Interval; Confidence Interval Reference; Credible Interval Reference; Distribution Record; Expected Value Record; Likelihood Category; Probability Transformation Record; Calibration-Relevant Probability Record; Probability Governance Gap; Uncertainty Governance Gap; Confidence Governance Gap; Representation Governance Gap; historical probability state; historical uncertainty state.

Internal Modules

PPUS is operationalized through five Core Modules:

1. PRPM — Probability Representation Module

Governs how numeric, qualitative, interval-based, distributional, and conditional probabilities are represented without semantic distortion.

2. PUNM — Predictive Uncertainty Module

Governs identification, classification, representation, propagation, concentration, interaction, and qualification of material predictive uncertainty.

3. CFRM — Confidence Representation Module

Governs the meaning, basis, representation, boundaries, and interpretation of predictive confidence independently from probability.

4. PRGM — Predictive Range Governance Module

Governs ranges, intervals, bounds, point estimates, distributional ranges, precision, rounding, and related representation.

5. CPM — Conditional Probability Module

Governs probability representations whose meaning depends upon assumptions, scenarios, states, events, or other predictive conditions.

The internal module flow is:

Probability Representation → Predictive Uncertainty → Confidence Representation → Predictive Range
Governance → Conditional Probability

Relationship to POPS

POPS establishes the Prediction Object, Purpose, Horizon, and Boundary.

PPUS preserves those constraints when probability and uncertainty are represented.

Relationship to PMES

PMES governs Prediction Methodology.

PPUS consumes the applicable methodology when determining what probability or uncertainty representation is legitimate.

Relationship to PBAS

PBAS governs Prediction Basis.

PPUS governs how limitations or uncertainty in that basis may affect predictive probability and uncertainty representation.

Relationship to PAS

PAS governs Predictive Assumptions.

PPUS governs probability and uncertainty conditional upon those assumptions where applicable.

Relationship to PFOS

PFOS establishes the specific Governed Prediction.

PPUS qualifies that prediction without silently changing its Predictive Claim.

Relationship to PFES

PFES governs Prediction Feasibility.

PPUS does not determine whether the predicted outcome can realistically occur.

Relationship to CPGS

CPGS governs Collective Prediction.

PPUS governs the probability semantics of collective outputs once legitimately formed.

Relationship to PCRS

PCRS governs communication and reliance.

PPUS establishes the probability, uncertainty, and confidence meaning PCRS must preserve.

Relationship to PMRS

PMRS governs monitoring and revision.

PPUS provides probability and uncertainty states that may later require reassessment.

Relationship to POFPS

POFPS governs outcome and performance.

It may later assess:

calibration; probability performance; error; predictive reliability.

PPUS does not itself determine performance.

Relationship to PRCS

PRCS governs complete historical continuity.

PPUS preserves the probability and uncertainty state applicable to each governed prediction version.

Critical Distinctions

Probability $\neq$ Certainty

Probability $\neq$ Confidence

Confidence $\neq$ Certainty

High Probability $\neq$ High Confidence

Low Probability $\neq$ Low Confidence

Same Probability Value $\neq$ Same Predictive Meaning

Probability Without Methodological Context $\neq$ Fully Governed Probability

Expected Value $\neq$ Most Probable Outcome

Mean $\neq$ Median $\neq$ Mode

Predictive Uncertainty $\neq$ Observed Prediction Error

Unquantified $\neq$ Nonexistent

Probability at Horizon A $\neq$ Probability at Horizon B

Conditional Probability Without Condition $\neq$ Same Probability

Scenario-Dependent Probability $\neq$ Unconditional Probability

Claim Precision $\neq$ Probability Precision

Confidence Interval $\neq$ Predictive Interval

Narrower Range $\neq$ Better Prediction by Default

Wide Range $\neq$ Poor Governance by Default

Numerical Precision $\neq$ Epistemic Precision

Calibration $\neq$ Individual Prediction Correctness

Likely $\neq$ Numerically Defined Probability by Default

Same Likelihood Label $\neq$ Same Probability Meaning

Score $\neq$ Probability by Default

Model Confidence $\neq$ Event Probability

Classification Confidence $\neq$ Predictive Probability

Market Price $\neq$ Governed Probability Estimate

Market-Implied Probability $\neq$ Governed Probability Estimate by Default

Average of Probabilities $\neq$ Governed Collective Probability by Default

Agreement Among Dependent Predictors $\neq$ Independent Confirmation

Simple Output $\neq$ Simple Uncertainty Structure

Prediction Formation $\neq$ Probability Determination

Probable $\neq$ Feasible

Feasible $\neq$ Probable

Strong Prediction Basis $\neq$ High Probability

Strong Assumption Support $\neq$ High Outcome Probability

Current Probability $\neq$ Historical Probability

Valid Initial Probability $\neq$ Valid Continuing Probability

Historical Predictive Conditions $\neq$ Current Predictive Conditions

Unknown Probability $\neq$ 50% Probability

Quantifiable $\neq$ Must Be Quantified

Probability Disagreement $\neq$ Governance Failure by Default

Probability Agreement $\neq$ Independent Confirmation by Default

Low Probability $\neq$ Low Consequence

Enterprise Application

PPUS allows organizations to use probability without allowing probability to become false certainty.

It is particularly relevant for:

AI forecasting systems; financial predictions; risk forecasting; healthcare forecasting; industrial maintenance predictions; weather and climate forecasting; election forecasting; prediction markets; supply-chain forecasting; project completion forecasting; autonomous decision systems.

Organizations frequently receive predictive outputs such as:

82%

High Confidence

Likely

90% confidence interval

without knowing whether these expressions describe the same thing.

PPUS forces their meanings apart.

This is essential because different probability representations can lead to radically different operational decisions even when they appear numerically similar.

Canonical Boundary

PPUS governs the representation, provenance, methodological association, conditionality, confidence, uncertainty, ranges, intervals, distributions, precision, transformation, qualitative likelihood, probability-state history, and semantic boundaries of probability and uncertainty associated with a Governed Prediction. It does not govern the formation of the Predictive Claim itself, Prediction Basis itself, Predictive Assumption identification itself, Prediction Feasibility, Collective Prediction composition, communication and reliance, prediction-level monitoring and revision, outcome determination, predictive performance assessment, general risk consequence, or Operational Reality determination.

PPUS governs:

how strongly a prediction may legitimately be expressed.

It does not establish:

whether the predicted future will occur.

Canonical Closing Statement

A probability number can make a prediction look precise long before the prediction is actually well understood.

PPUS separates probability, confidence, uncertainty, range, conditionality, and provenance so that predictive strength cannot be manufactured by notation alone.

A prediction may carry a probability. It must never inherit certainty simply because the probability has been quantified.

Nahlad

Prediction Probability & Uncertainty Standard

Architecture:

PGA™ — Prediction Governance Architecture

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Canonical Definition: The Prediction Probability & Uncertainty Standard governs how probability, uncertainty, confidence, ranges, conditionality, and related predictive qualification are represented in association with a Governed Prediction without overstating what the prediction actually establishes.

Governed Space:

Predictive Probability & Uncertainty

Core Question: How should probability, uncertainty, confidence, ranges, and conditionality be represented without overstating what the prediction actually establishes?

Probability ≠ Confidence ≠ Certainty

A quantified prediction must not inherit certainty merely because its uncertainty has been expressed numerically.

→ View Standard

About standard

Governed Links

[About Prediction Probability & Uncertainty Standard \(PPUS\).](#)

[Prediction Probability Representation Module \(PRPM\).](#)

[Prediction Uncertainty Module \(PUNM\).](#)

[Confidence Representation Module \(CFRM\).](#)

[Prediction Range Governance Module \(PRGM\).](#)

[Conditional Probability Module \(CPM\).](#)

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